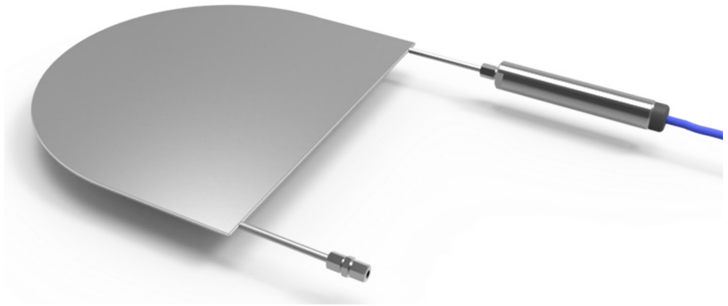
Methods of In-situ Stress Measurements		
	Method	**Volume (m^3)**
Hydraulic Methods	Hydraulic fracturing	0.5-50
	Sleeve Fracturing	10^{-2}
	Hydraulic tests on pre-existing fractures HTPF	1-10
Relief Methods	Surface relief methods	1-2
	Undercoring	10^{-3}
	Borehole relief method (overcoring BH,slotting etc.)	$10^{-3} – 10^{-2}$
	Relief of a large rock volumes (bored raise , under excavation technique etc.	$10^{2} – 10^{3}$
Jacking Method	Flat jack method	0.5 – 2
	Curved jack method	10^{-2}
Strain Recovery Method	Anelastic strain Recovery (ASR)	10^{-3}
	Differential strain curve analysis (DSCA)	10^{-4}
Borehole Breakout Method	Caliper and dipmeter analysis	$10^{-2} – 10^{2}$
	Borehole televiewer analysis	$10^{-2} – 10^{2}$
Other methods	Fault Slip Data Analysis	10^{8}
	Earthquake focal mechanism	10^{9}
	Indirect Methods (Kaiser Effect etc.)	$10^{-4} – 10^{-3}$
	Inclusion in time dependent rocks	$10^{-2} – 1$
	Measurement of residual stresses	$10^{-5} – 10^{-3}$

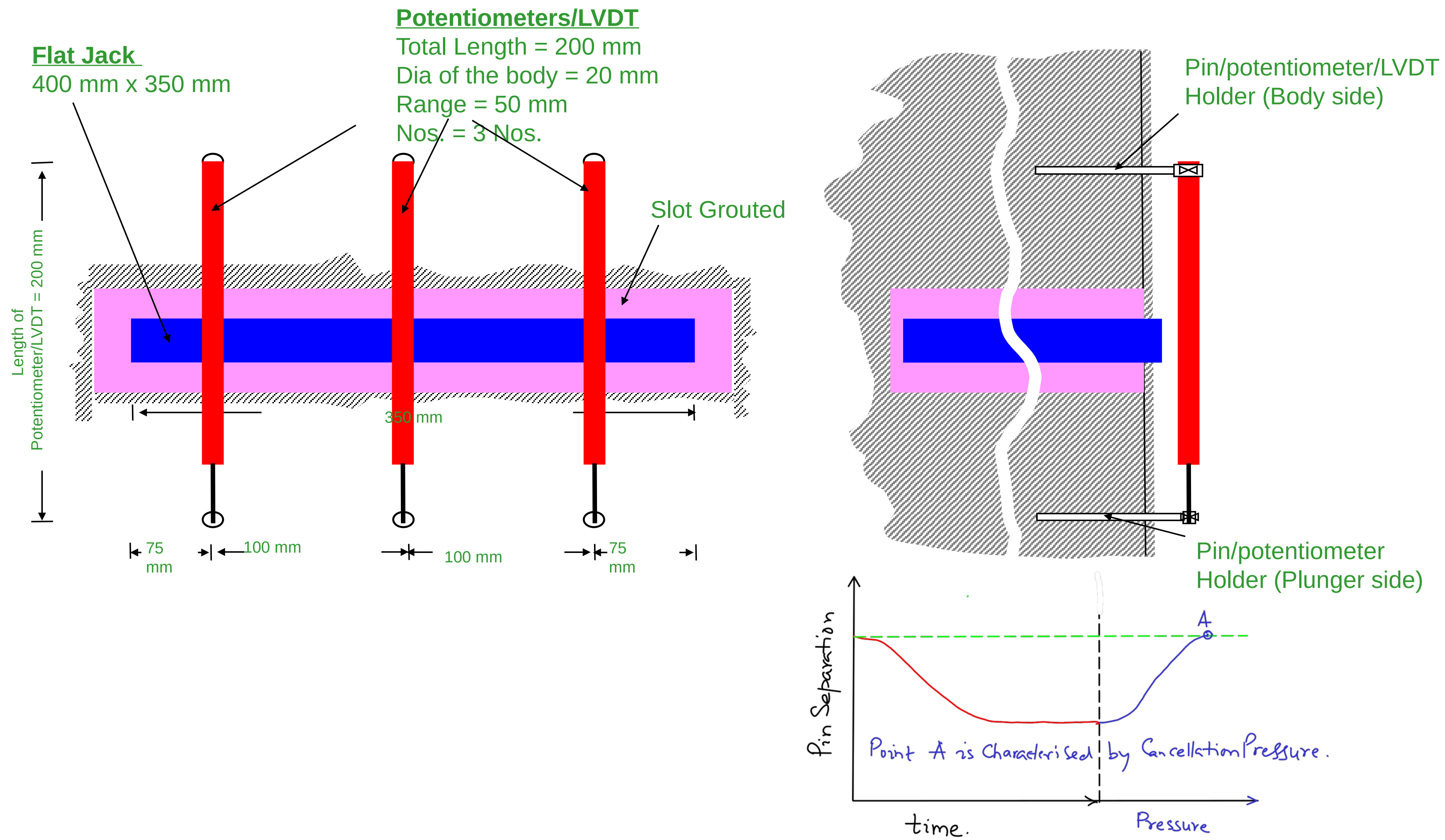
Flat jack testing

Determination of in-situ stress state at accessible surfaces (one normal stress component per slot). Typical applications include:

- In buildings: actual loads on pillars and footings
- In tunnels: actual loads on tunnel linings
- In dam abutments: in-situ rock stresses, often carried out as

part of Large Flat Jack testing.




Potentiometers/LVDT
Total Length = 200 mm
Dia of the body = 20 mm
Range = 50 mm
Nos. = 3 Nos.
Flat Jack
400 mm x 350 mm
Length of Potentiometer/LVDT = 200 mm
Slot Grouted
350 mm
75 mm
100 mm
100 mm
75 mm
Pin/potentiometer/LVDT Holder (Body side)
Pin/potentiometer Holder (Plunger side)
Pin Separation
A
Point A is Characterised by Cancellation Pressure.
time.
Pressure

The measurement consists of two distinct phases:

- Cutting a slot (preferably by a diamond saw)
- Insertion of a flat jack into the slot and pressurisation until Compensation (= reestablishing the deformation stage prior to saw cutting)

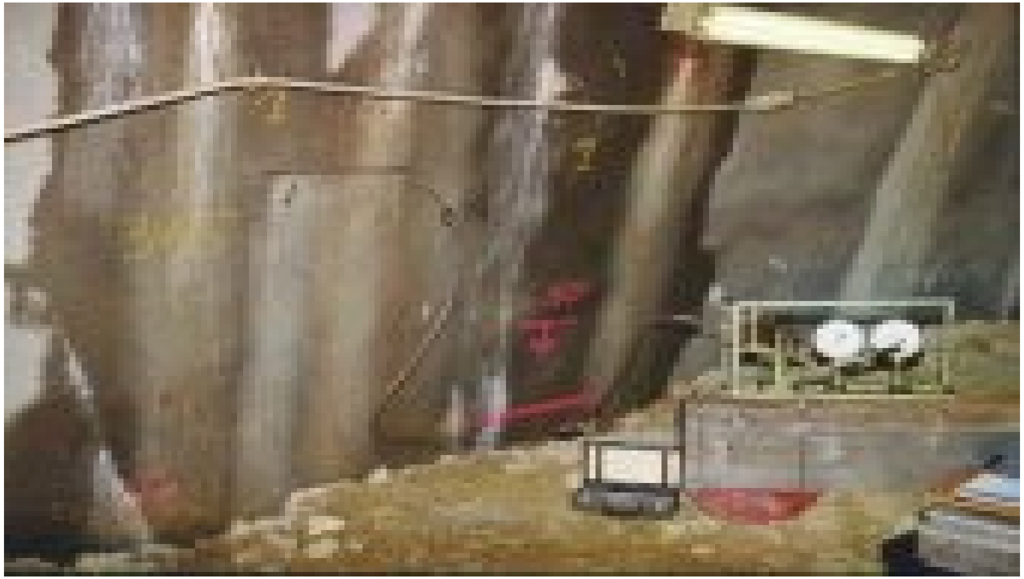
$\sigma_n = \rho * K_m * K_a$ with:

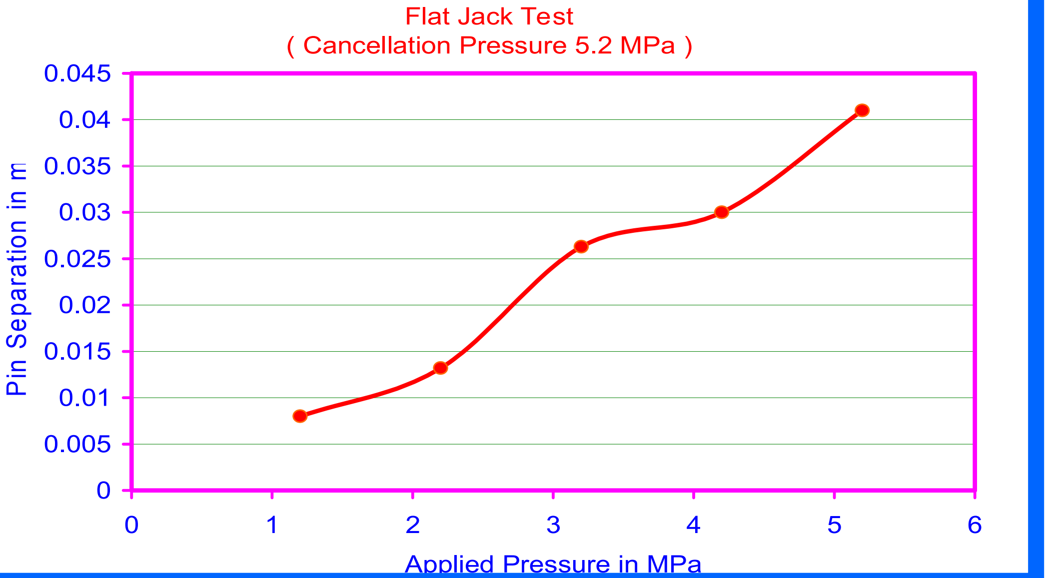
σn = normal stress component to be determined

ρ = hydraulic pressure in jack at full compensation

K_m = shape factor of the jack

K_a = proportion of area of jack to area of slot



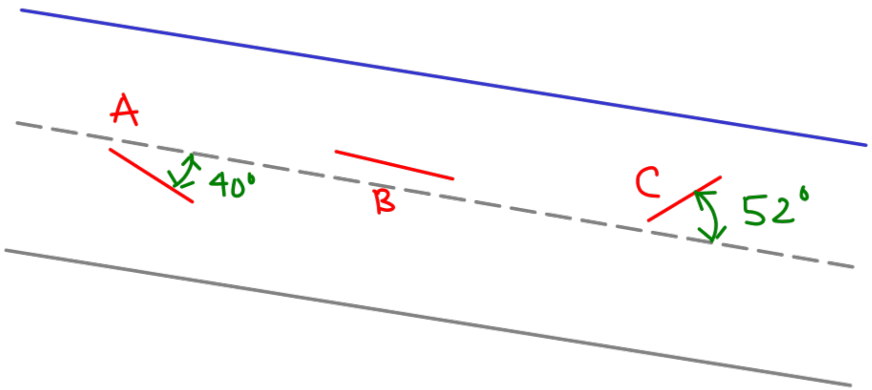


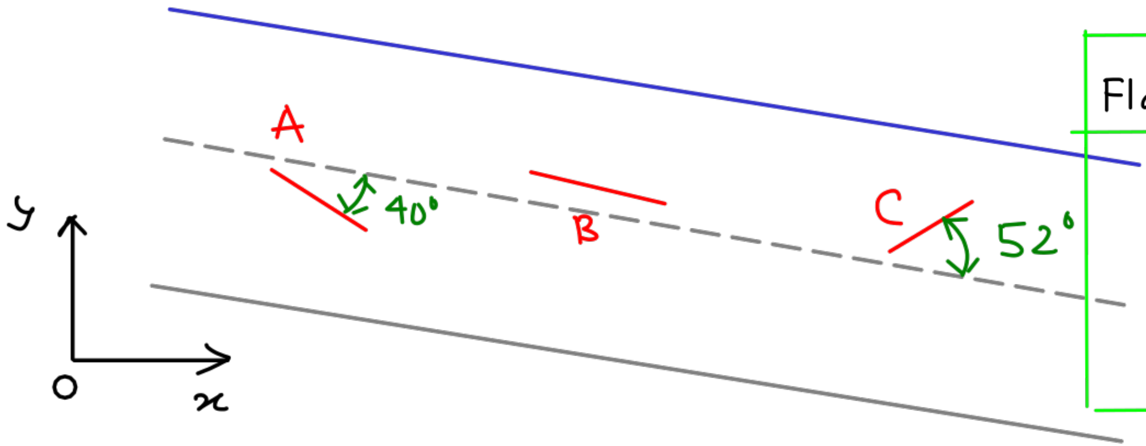
Flat Jack Test
(Cancellation Pressure 5.2 MPa)
Pin Separation in m
0.045
0.04
0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
0
1
2
3
4
5
6
Applied Pressure in MPa

The particular advantage of the flat jack method is that knowledge of the elastic constants of the tested material (e.g. Young's Modulus E and Poisson's ratio (ν) is not required. This is in contrast to other stress measuring methods.

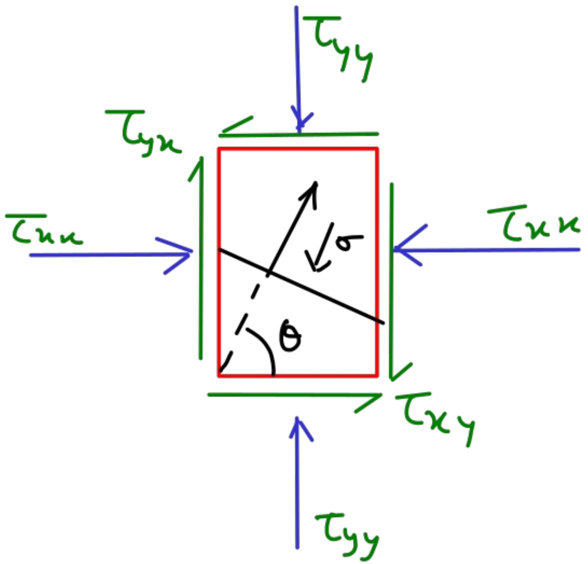
Problem:-

Three flat jack tests have been made close to eachother in the wall of a long straight tunnel, the axis of which dips at 7°. The measurement site is approximately 250 m below the ground surface. (it is assumed that the flat jacks operate in the same stress field. The slots for the flat jacks are cut normal to the wall of the tunnel and are oriented relative to the tunnel axis as shown. The cancellation pressure for each of the flat-jacks A, B & C is 7.56 MPa, 6.72 MPa and 7.5 MPa respectively. Compute the principal stresses & their direction & ascertain whether they are in accordance with the world-wide trend.





Flat Jack	F.J angle w.r.t x-axis	Angle Normal To FJ.
A	$40° + 7°$	$90° - 47° = 43°$
B	$7°$	$90 - 7 = 83$
C	$-52° + 7°$	$90 + 52 - 7 = 135$

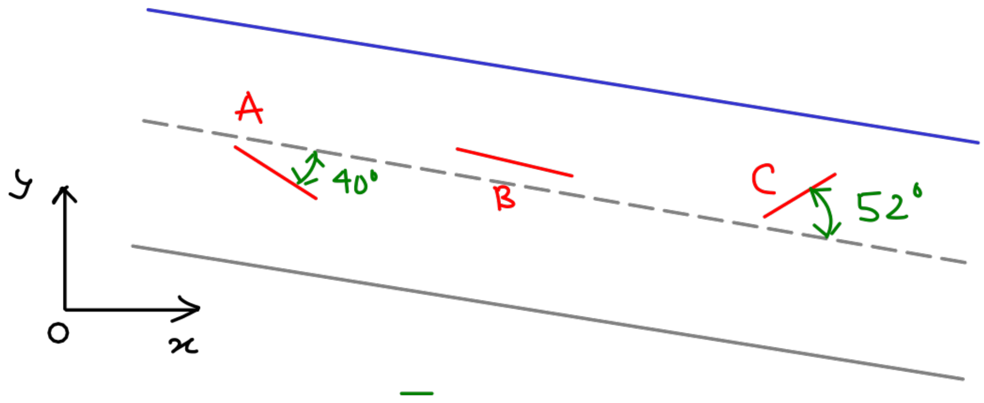


$$\tau' = L \tau L^T \qquad L = \begin{bmatrix} c & s \\ -s & c \end{bmatrix}$$

$$\begin{bmatrix} c & s \\ -s & c \end{bmatrix} \begin{bmatrix} \tau_{xx} & \tau_{xy} \\ \tau_{xy} & \tau_{yy} \end{bmatrix} \begin{bmatrix} c & -s \\ s & c \end{bmatrix} = \begin{bmatrix} c\tau_{xx} + s\tau_{xy} & c\tau_{xy} + s\tau_{yy} \\ \checkmark & \checkmark \end{bmatrix} \begin{bmatrix} c & -s \\ s & c \end{bmatrix}$$

$$= \begin{bmatrix} c[c\tau_{xx} + s\tau_{xy}] + s[c\tau_{xy} + s\tau_{yy}] & \checkmark \\ \checkmark & \checkmark \end{bmatrix}$$

$$\Rightarrow \quad \tau_{x'x'} = \sigma = c^2\tau_{xx} + s^2\tau_{yy} + 2cs\,\tau_{xy}$$



$$\sigma_\theta = c^2\tau_{xx} + s^2\tau_{yy} + 2cs\,\tau_{xy}$$

$$7.56 = \tau_{xx}\cdot\cos^2 43 + \tau_{yy}\sin^2 43 + 2\tau_{xy}\sin 43\cos 43$$

$$6.72 = \tau_{xx}\cdot\cos^2 83 + \tau_{yy}\sin^2 83 + 2\tau_{xy}\sin 83\cos 83$$

$$7.5 = \tau_{xx}\cdot\cos^2 135 + \tau_{yy}\sin^2 135 + 2\tau_{xy}\sin 135\cos 135$$

$$\begin{bmatrix}\sigma_{\theta_1}\\ \sigma_{\theta_2}\\ \sigma_{\theta_3}\end{bmatrix} = [R]_{3\times 3}\begin{bmatrix}\tau_{xx}\\ \tau_{yy}\\ \tau_{xy}\end{bmatrix} \quad \text{or.} \quad R^{-1}\begin{bmatrix}\sigma_{\theta_1}\\ \sigma_{\theta_2}\\ \sigma_{\theta_3}\end{bmatrix} = \begin{bmatrix}\tau_{xx}\\ \tau_{yy}\\ \tau_{xy}\end{bmatrix}$$

$$\begin{bmatrix}\tau_{xx}\\ \tau_{yy}\\ \tau_{xy}\end{bmatrix} = \begin{bmatrix}\cos 43^\circ & \sin^2 43 & 2\sin 43\cos 43\\ \cos 83^\circ & \sin^2 83 & 2\sin 83\cos 83\\ \cos^2 135^\circ & \sin^2 135^\circ & 2\sin 135\cos 135\end{bmatrix}^{-1}\begin{bmatrix}7.56\\ 6.72\\ 7.5\end{bmatrix} = \begin{bmatrix}8.308\\ 6.696\\ 0.002\end{bmatrix}$$

Problem:

Further to the experiments as above, two more slots are created with dip of slot axis 20° & 90° anticlockwise relative to the tunnel axis and produce cancellation pressures of 7.38 MPa and 7.86 MPa; respectively. Compute the best estimate of the principal stress.

This problem can be generalised to

$$\begin{bmatrix} \sigma_{\theta 1} \\ \sigma_{\theta 2} \\ \vdots \\ \sigma_{\theta n} \end{bmatrix} = \begin{bmatrix} \cos^2\theta_1 & \sin^2\theta_1 & 2\sin\theta_1\cos\theta_1 \\ \cos^2\theta_2 & \sin^2\theta_2 & 2\sin\theta_2\cos\theta_2 \\ \vdots & \vdots & \vdots \\ \cos^2\theta_n & \sin^2\theta_n & 2\sin\theta_n\cos\theta_n \end{bmatrix} \begin{bmatrix} \tau_x \\ \tau_y \\ \tau_{xy} \end{bmatrix}$$

$$[\sigma_{FJ}]_{n\times 1} = [R]_{n\times 3}\,[\tau]_{3\times 1}$$

Regression Ref:- Numerical Recipes in Fortran — by W.H. Press 2nd edition

$$[\sigma_{FJ}]_{n\times1} = [R]_{n\times3}\,[\tau]_{3\times1}$$

Pre-multiply both sides by $[R]^T$

$$\Rightarrow [R^T]_{3\times n}\,[\sigma_{FJ}]_{n\times1} = [R^T]_{3\times n}\,[R]_{n\times3}\,[\tau]_{3\times1}$$

$$= [R^TR]_{3\times3}\,[\tau]_{3\times1}$$

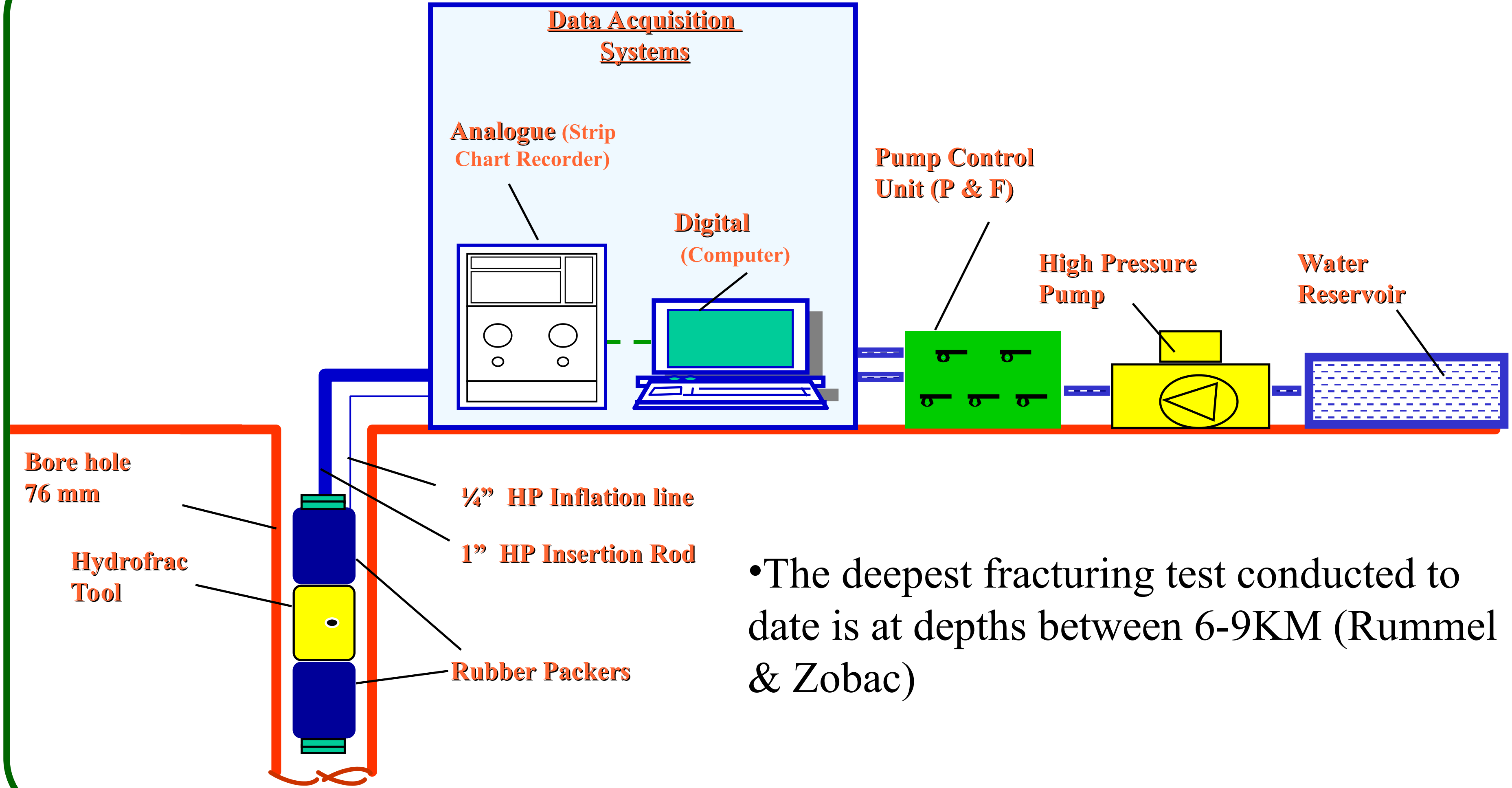
$$\text{or, } [R^TR]^{-1} R^T \sigma_{FJ} = \underbrace{[R^TR]^{-1}\cdot[R^TR]}_{I}\,[\tau]$$

$$\text{or, } [\tau] = (R^TR)^{-1} R^T \sigma_{FJ}.$$

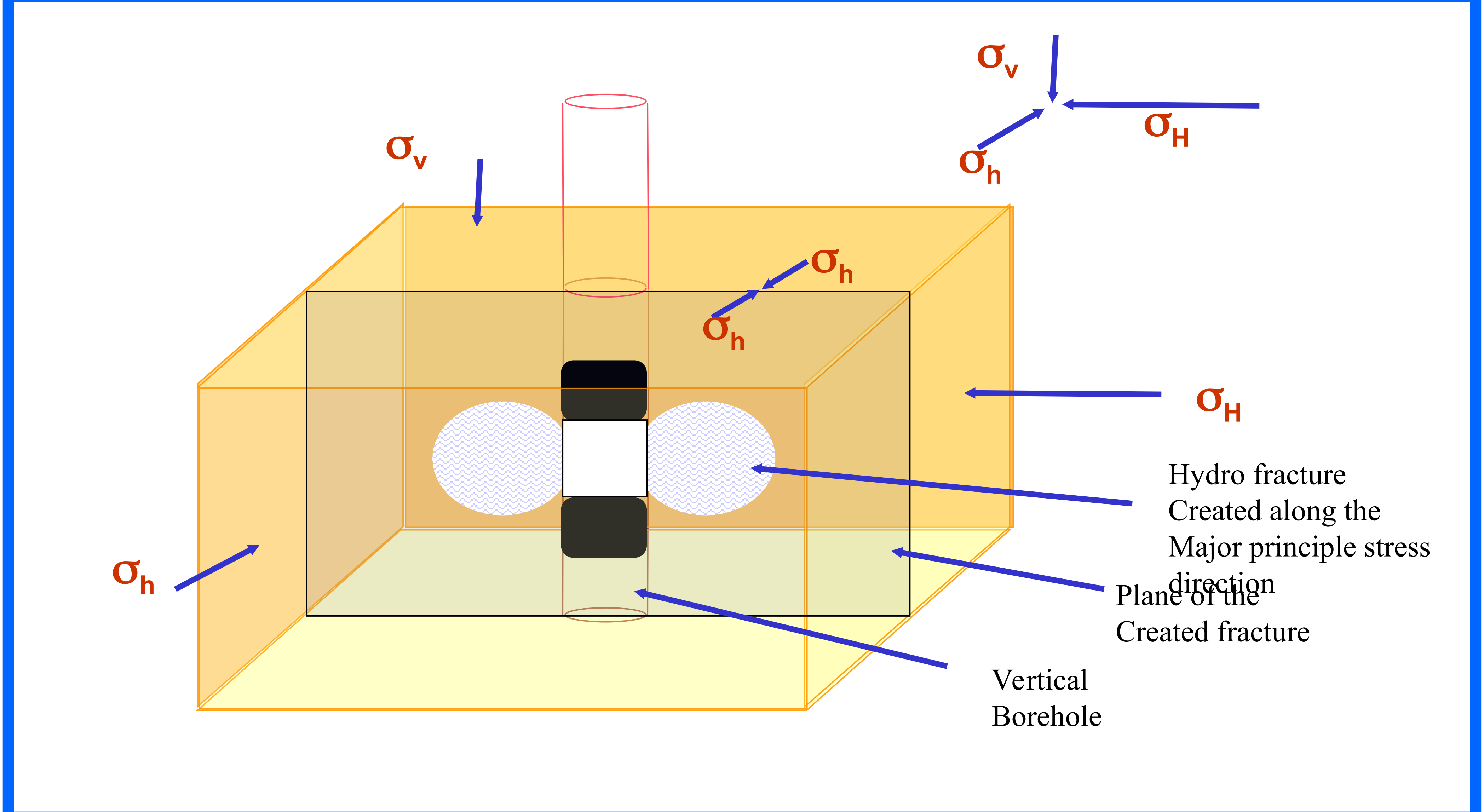
For our Example with 5-tests

$$\begin{bmatrix}\tau_x\\ \tau_y\\ \tau_{xy}\end{bmatrix} = \begin{bmatrix}7.99\\ 6.90\\ 0.07\end{bmatrix}\text{MPa.}$$

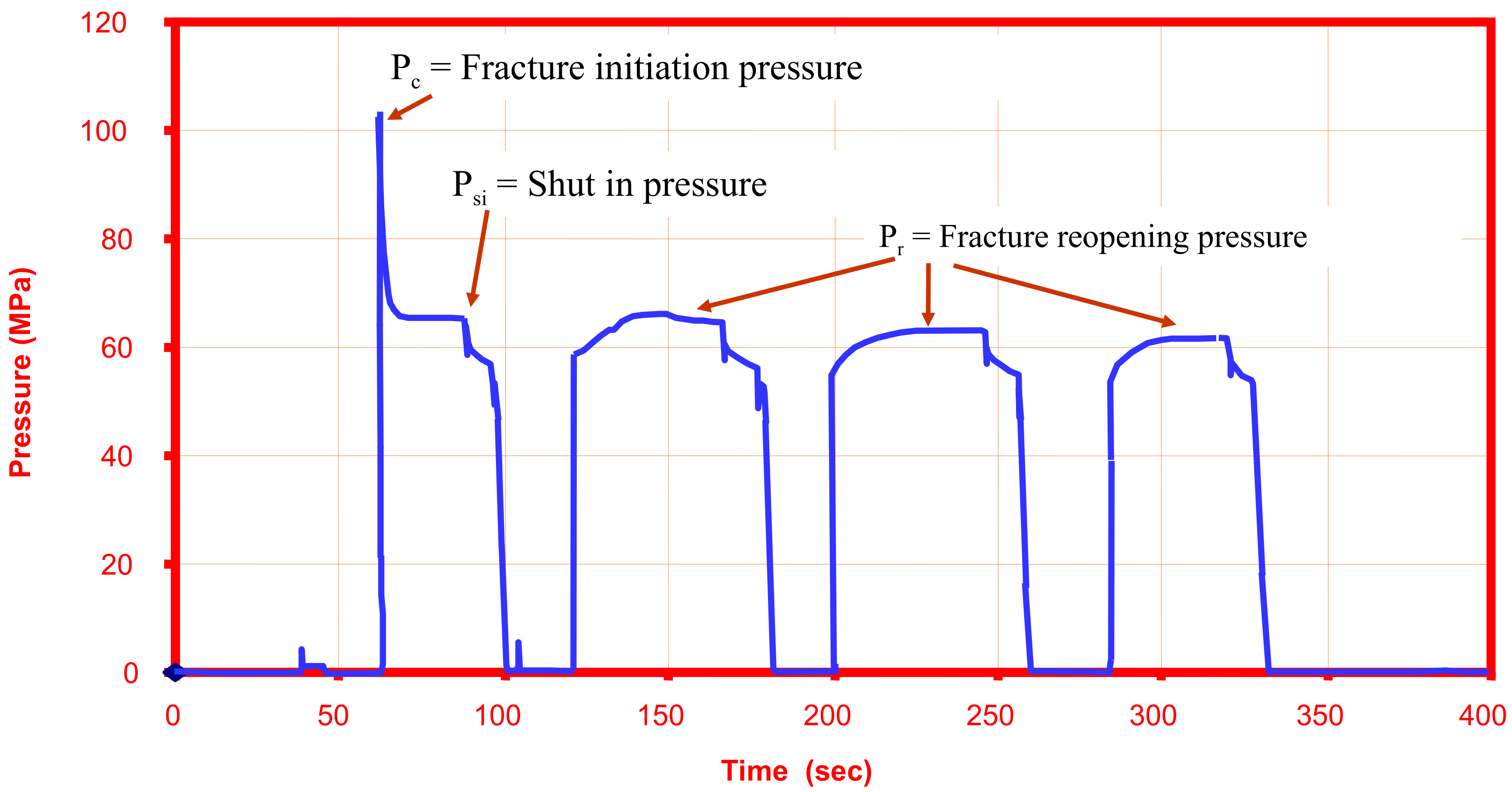
Pl. Calculate & report.



- The deepest fracturing test conducted to date is at depths between 6-9KM (Rummel & Zobac)



σ_v
σ_H
σ_h
σ_v
σ_h
σ_h
σ_H
σ_h
Hydro fracture Created along the Major principle stress direction
Plane of the Created fracture
Vertical Borehole



P_c = Fracture initiation pressure
P_{si} = Shut in pressure
P_r = Fracture reopening pressure
Pressure (MPa)
120
100
80
60
40
20
0
0
50
100
150
200
250
300
350
400
Time (sec)

Classical Theory of Hydraulic Fracturing

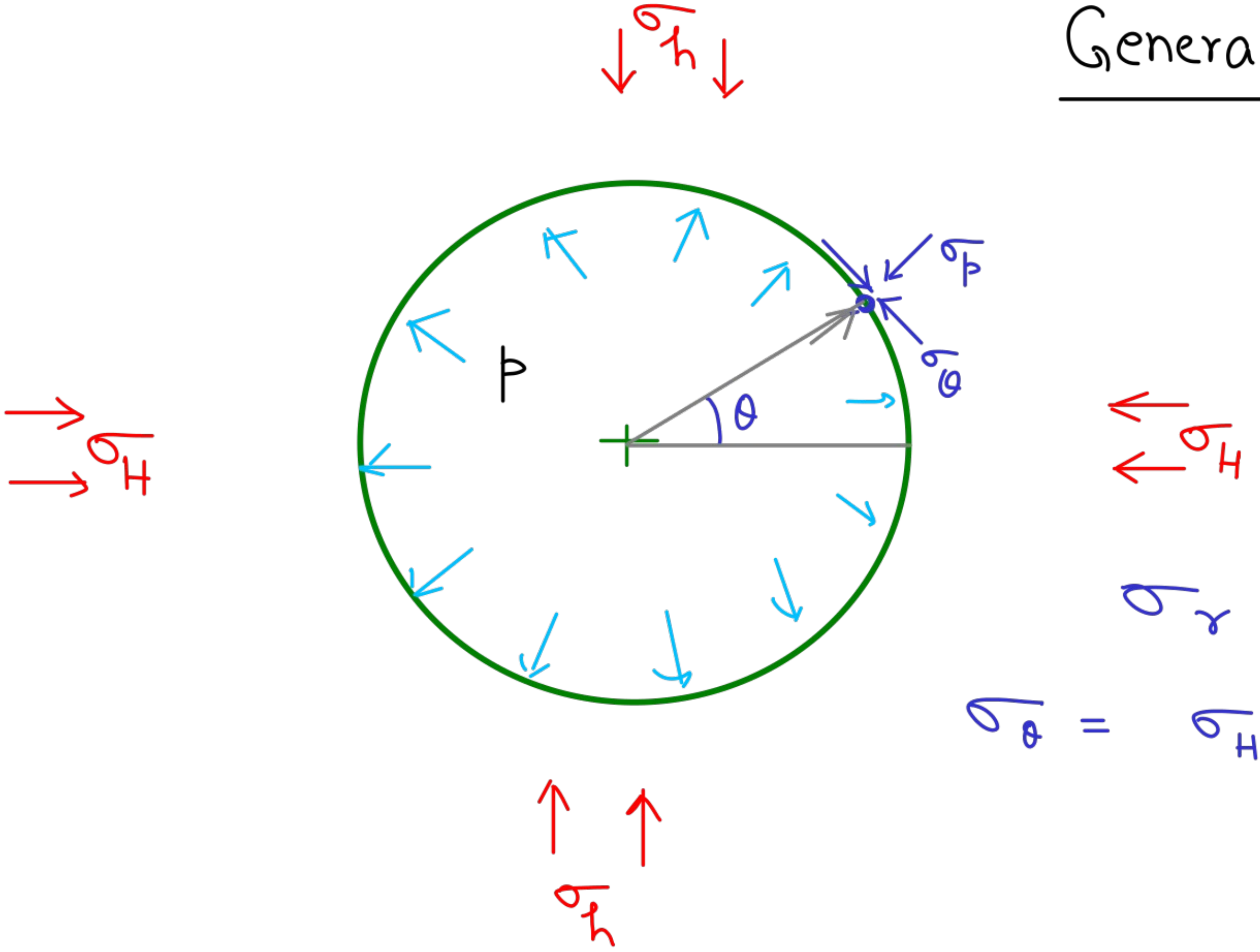
Hubbert & Willis (1957)

Assumptions:

1. The rock mass is CHILE & impermeable to fluid.
2. The borehole is to be drilled $\|^{el}$ to one of the principal stresses
3. Fracture initiates when

   Net tangential stress on BH wall [induced by far field stress & fluid Pr] = Tensile strength of wall rock.

4. The fracture initiates and propagates in direction of least resistant path (corallary of the MoS)

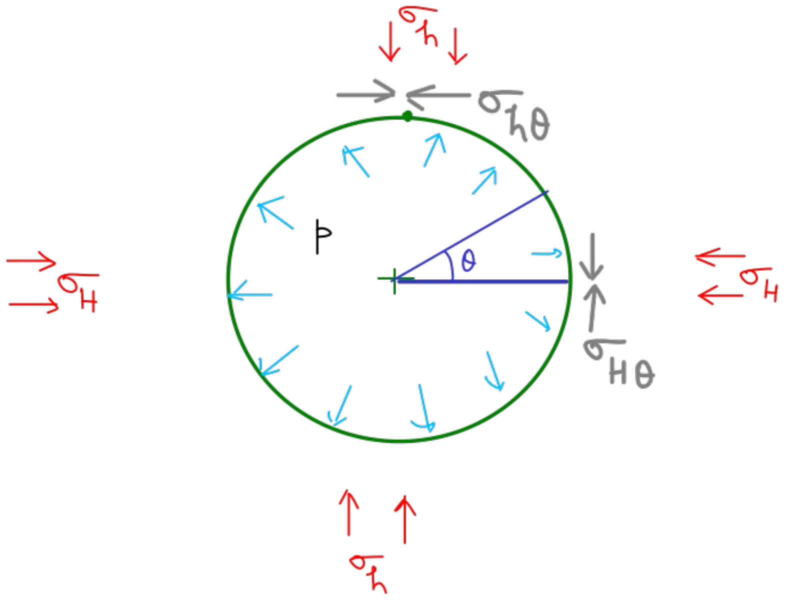


General Elastic Solution

Vertical BH
- Far field Stresses (σ_H & σ_h)
- internal pressure p

$$\sigma_r = p$$

$$\sigma_\theta = \sigma_H + \sigma_h - p - 2(\sigma_H - \sigma_h)\cos 2\theta$$



$$\sigma_r = p$$

$$\sigma_\theta = \sigma_H + \sigma_h - p - 2(\sigma_H - \sigma_h)\cos 2\theta$$

$$\forall\ \theta = 0^\circ \quad \sigma_{H\theta} = (3\sigma_h - \sigma_H) - p \quad \text{——} \quad 1$$

$$\forall\ \theta = 90^\circ \quad \sigma_{h\theta} = (3\sigma_H - \sigma_h) - p \quad \text{——} \quad 2$$

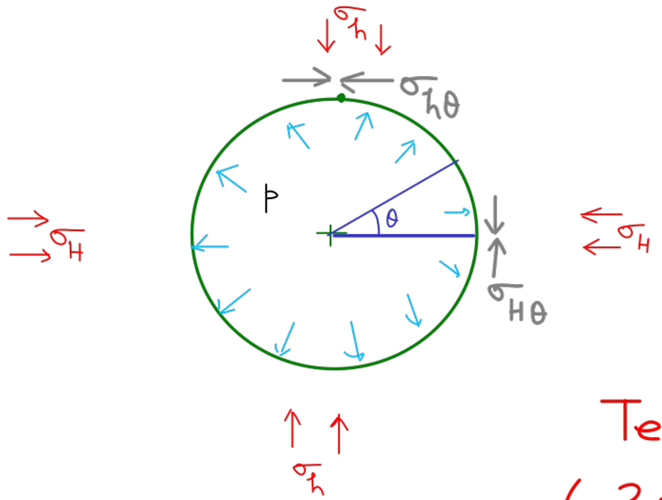
Since $\sigma_h < \sigma_H$ (By defn)

$$4\sigma_h < 4\sigma_H$$

$$(3\sigma_h - \sigma_H) < (3\sigma_H - \sigma_h)$$

$$(3\sigma_h - \sigma_H) - p < (3\sigma_H - \sigma_h) - p$$

$$\therefore\ \sigma_{H\theta} < \sigma_{h\theta}.$$



$\sigma_{H\theta} < \sigma_{h\theta} \Rightarrow \sigma_{H\theta}$ has least magnitude.

As long as $\sigma_{H\theta} > 0$, it is compressional.

or, $(3\sigma_h - \sigma_H) - p > 0$ [compressional]

As pressure ↑ $3\sigma_h - \sigma_H < 0$ [Tensile]

Tensile stress will be formed when

$$(3\sigma_h - \sigma_H) - p_c = -\sigma_t = -T$$

The pressure at which fracture initiates is the Critical pressure (p_c)

$$\therefore \; p_c = (3\sigma_h - \sigma_H) + T \qquad (3)$$

In the next refrac cycle (as fracture is already created). The pressure (p_r) required to open the fracture is just a limiting Condition.

$$p_r = (3\sigma_h - H) \qquad (4)$$

From (3) & (4) $\boxed{p_c - p_r = T}$

The stress counteracting the fracture is nothing but Shut-in-pressure.

or $P_{si} = \sigma_h$

Problem:-

In a hydraulic fracturing test in gneiss, an induced fracture was created at 12.8 MPa. The fracture was re-opened in subsequent cycles. The pore-water pressure is negligible. The re-opening pressure in third cycle was 6.4 MPa.

Determine the maximum principal horizontal stress if the shut-in pressure was 5.9 MPa.

<u>Given:-</u>

$p_c = 12.8 \text{ MPa}.$

$p_r = 6.4 \text{ MPa}$

$p_{sh} = 5.9 \text{ MPa}.$

Using Eqn(3) & (4) $T = p_c - p_r = 12.8 - 6.4 = 6.4 \text{ MPa}.$

$$p_c = (3\sigma_h - \sigma_H) + T$$

$$12.8 = 3 \times 5.9 - \sigma_H + 6.4$$

$$\Rightarrow \boxed{\sigma_H = 11.3 \text{ MPa}}$$

HTPF Method :

• proposed by Cornet and Valette (1984).

•It is the Fracture Pressurization Method, originally called 'hydraulic tests on preexisting fractures' (HTPF) by Cornet (1986).

•By using the general theory of stress in 3-dimensional space, the fracture-normal stress can be formulated as a nonlinear function of depth and fracture orientation, expressed in the form of 6 unknown parameters which uniquely define the stress tensor.

• Unlike the conventional models, the fracture pressurization method **neither invokes the stress-strain relations nor assumes the idealistic (homogeneous, isotropic, and linear elastic) rock properties.**


Analog Data Acquisition system
Pentium Computer for digital Acquisition

Hydrofrac Tool Inserted inside one of the Vertical Boreholes

Hydrofrac Tool Inserted inside one of the Horizontal Boreholes

Hydrofrac tool inserted inside one of NX Boreholes

Rope and Pulley used to insert the Hydrofrac tool insidethe NX size borehole,


Hydrofrac tool

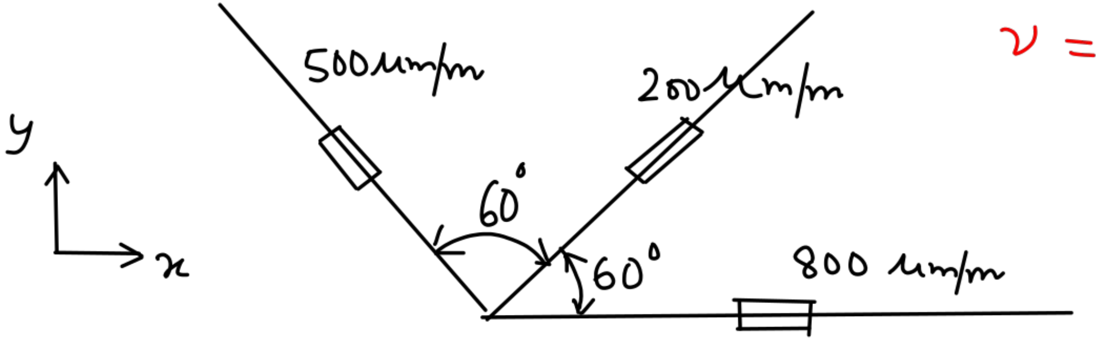
Measuring strains to deduce stress

Measurement of principal stress is conducted along a Vertical plane with 60° Strain rosette. The normal Strains are obtained as given in figure below.

Given:-

$E = 5 GPa$

$\nu = 0.25$



Determine the magnitude of principal stress and the direction of principal stresses w.r.t x-axis.

Since Strain is also a 2^{nd} order tensor, all mathematical operations will be similar.

$$\forall \text{ Stress} \quad \sigma_\theta = \tau_{xx}\cos^2\theta + \tau_{yy}\sin^2\theta + 2\tau_{xy}\sin\theta\cos\theta.$$

Analogously $$\epsilon_\theta = \epsilon_{xx}\cos^2\theta + \epsilon_{yy}\sin^2\theta + 2\epsilon_{xy}\sin\theta\cdot\cos\theta$$

$\therefore$ For 3-measurements
$$\begin{bmatrix}\epsilon_{\theta_1}\\ \epsilon_{\theta_2}\\ \epsilon_{\theta_3}\end{bmatrix} = \begin{bmatrix}\cos^2\theta_1 & \sin^2\theta_1 & 2\sin\theta_1\cos\theta_1\\ \cos^2\theta_2 & \sin^2\theta_2 & 2\sin\theta_2\cos\theta_2\\ \cos^2\theta_3 & \sin^2\theta_3 & 2\sin\theta_3\cos\theta_3\end{bmatrix}\begin{bmatrix}\epsilon_{xx}\\ \epsilon_{yy}\\ \epsilon_{xy}\end{bmatrix}$$

w.r.t x-y System

Given:
$\theta_1 = 0$ $\quad \epsilon_{\theta_1} = 800$
$\theta_2 = 60°$ $\quad \epsilon_{\theta_2} = 200$
$\theta_3 = 120°$ $\quad \epsilon_{\theta_3} = 500$

$$\Rightarrow [\epsilon_\theta] = [R][\epsilon]$$
$$\text{or } [\epsilon] = R^{-1}\epsilon_\theta$$
$$\Rightarrow \begin{bmatrix}\epsilon_{xx}\\ \epsilon_{yy}\\ \epsilon_{xy}\end{bmatrix} = \begin{bmatrix}800\\ 200\\ -173.2051\end{bmatrix}.$$

Finding the principal strain is eigen Value problem

$$\epsilon = \begin{bmatrix} 800 & -173.2051 \\ -173.2051 & 200 \end{bmatrix} \qquad \begin{vmatrix} 800-\epsilon & -173.2051 \\ -173.2051 & 200-\epsilon \end{vmatrix} = 0.$$

or, $$\epsilon^2 - 1000\epsilon + 160000 - 173.2051^2 = 0$$

$$\epsilon_1 = 846.4102 \ \mu m/m$$

$$\epsilon_2 = 153.5898 \ \mu m/m$$

Hooke's law [in principal co-ordinates]

Strains due to σ_{xx}: $\epsilon_{xx} = \dfrac{\sigma_{xx}}{E}$, $\epsilon_{yy} = -\nu \dfrac{\sigma_{xx}}{E}$

Strains due to σ_{yy}: $\epsilon_{xx} = \dfrac{-\nu\sigma_{yy}}{E}$, $\epsilon_{yy} = \dfrac{\sigma_{yy}}{E}$.

$$\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \end{bmatrix} = \frac{1}{E}\begin{bmatrix} 1 & -\nu \\ -\nu & 1 \end{bmatrix}\begin{bmatrix} \sigma_1 \\ \sigma_2 \end{bmatrix}$$

or, $$\begin{bmatrix} \sigma_1 \\ \sigma_2 \end{bmatrix} = \frac{E}{1-\nu^2}\begin{bmatrix} 1 & \nu \\ \nu & 1 \end{bmatrix}\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \end{bmatrix}$$

$$\sigma_1 = \frac{E}{1-\nu^2}(\epsilon_1 + \nu\epsilon_2)$$

$$\sigma_2 = \frac{E}{1-\nu^2}(\epsilon_2 + \nu\epsilon_1).$$

$$\sigma_1 = \frac{5\times10^9}{1-0.25^2}[846.4102 + 0.25\times153.5898]\times10^{-6}\ Pa = 4718.97\times10^3 Pa = 4.72\ MPa.$$

$$\sigma_2 = 1.95\ MPa.$$

Finding the Strain Direction is Eigen Vector Problem:

Direction of principal Strain = Direction of principal Stress

Substitute ϵ_1 in matrix.

$$\begin{bmatrix} 800-846.4102 & -173.2051 \\ -173.2051 & 200-846.4102 \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -46.4102 & -173.2051 \\ -173.2051 & -646.4102 \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} 1 & 173.2051/46.4102 \\ 1 & +\frac{646.4102}{173.2051} \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \end{bmatrix} = 0 \quad \text{or} \quad \begin{bmatrix} 1 & 3.73205 \\ 1 & 3.73205 \end{bmatrix} \begin{bmatrix} n_1 \\ n_2 \end{bmatrix} = 0$$

$$n_1 = 3.73205\, n_2 .$$

The eigen vector or the direction of σ_1 is $\begin{bmatrix} 3.73205 \\ 1 \end{bmatrix}$

Unit vector along σ_1 direction $= \begin{bmatrix} 0.96572 \\ 0.25957 \end{bmatrix}$

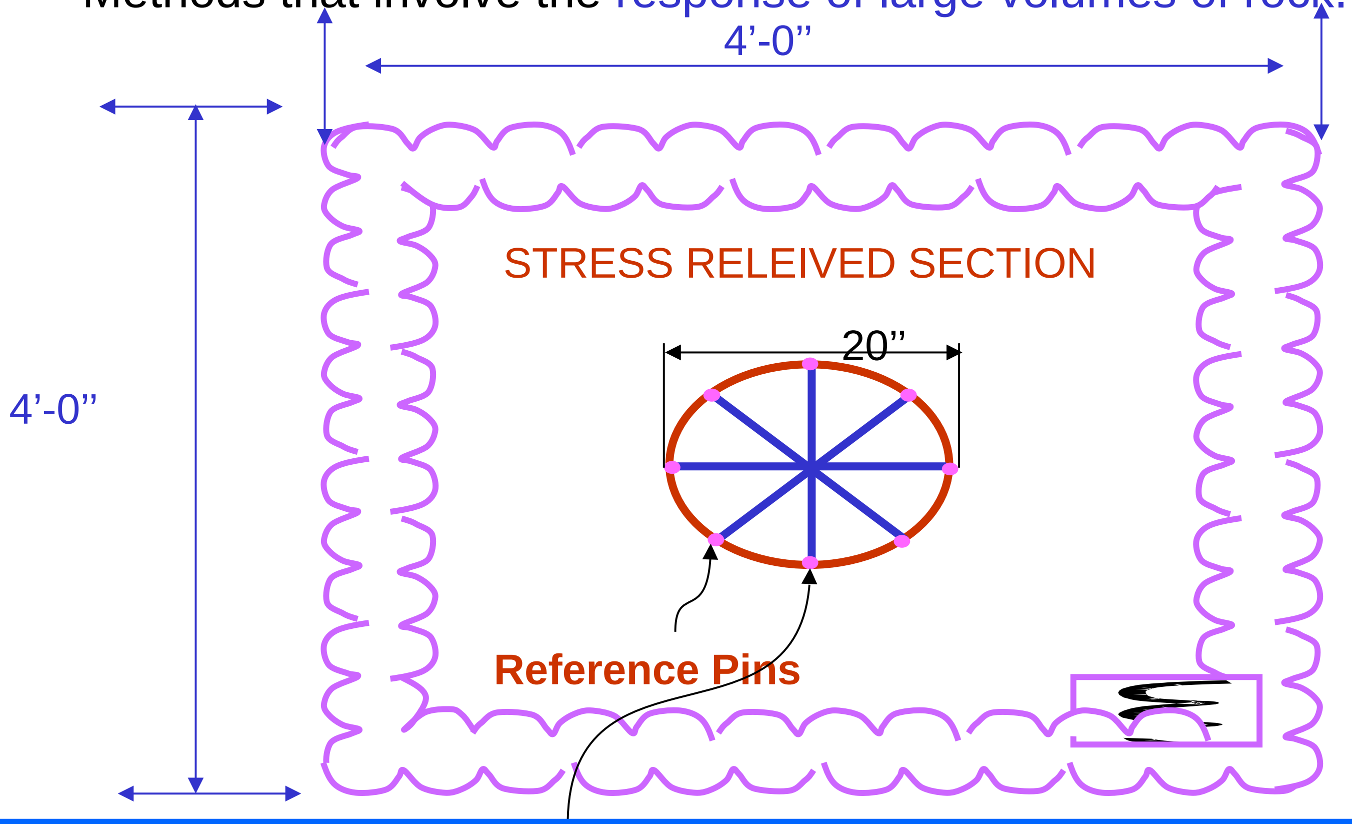
$\therefore$ σ_1 direction makes an angle of $\cos^{-1}(0.96572)$ w.r.t x-axis.

or, σ_1 direction makes an angle of 15.045^0 w.r.t. x-axis.

$\sigma_2 \perp^r$ to σ_1 $\Rightarrow$ σ_2 makes an angle of 105.045^0 w.r.t x-axis.

Relief Methods :

- •Methods that involve strain or displacement measurements on rock surface.
- •The methods that use instruments in boreholes
- •Methods that involve the response of large volumes of rock.



Lieurance Method (1933)

Rock mass relief methods :

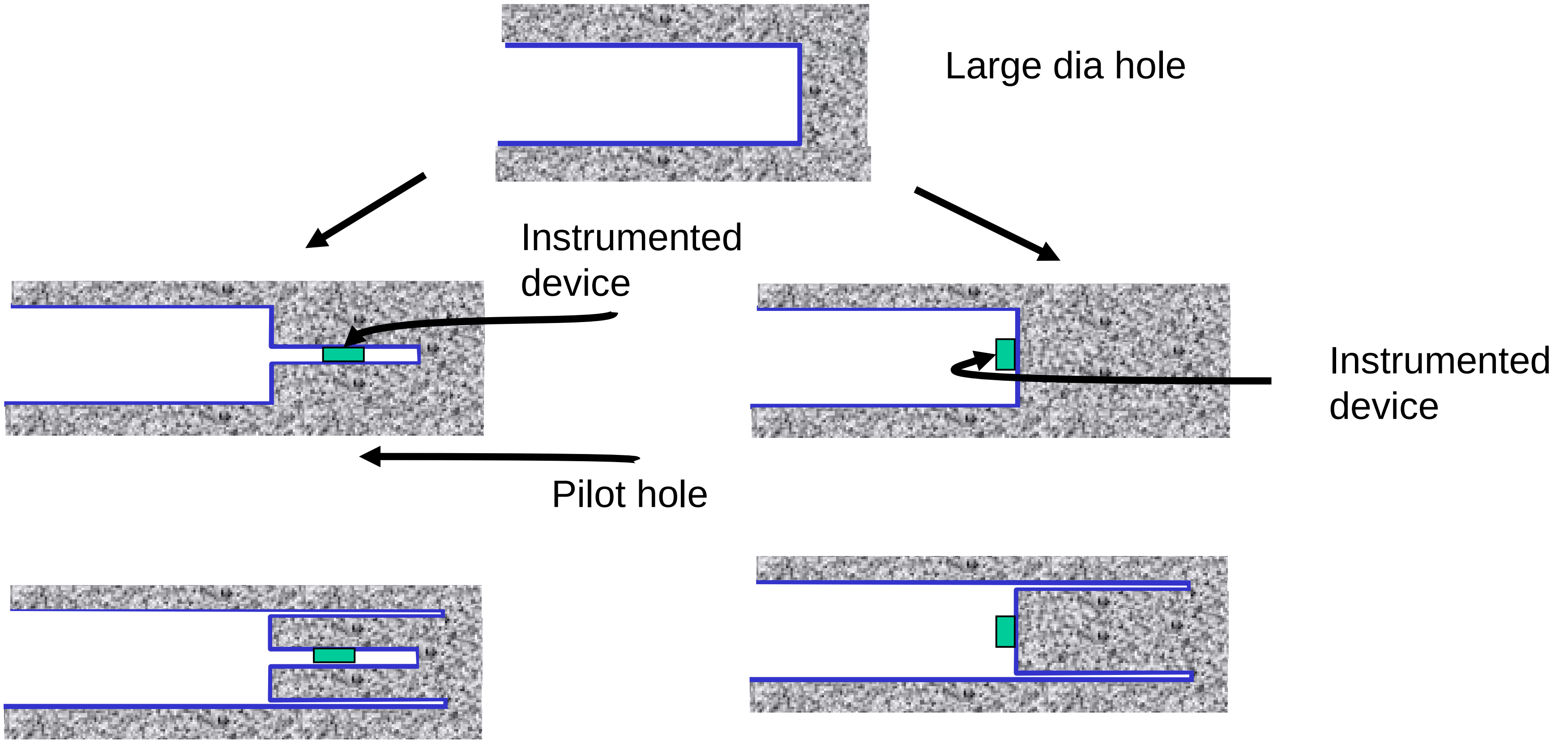
Reported by Brady et.al.(1976)

- They over-cored a large no. of strain rosettes glued on a wall of 1.8m bored raise at Mount ISA mine.
- Each rosette consisted of 10 measuring pins arranged in 5-diametric pairs of the circumference of a 250mm dia circle.
- The rosettes were over-cored with a 360mm thin walled diamond bit.
- A total of 14 measurement stations were established at different elevations in the bored raise.
- At each station, four rosettes were installed and five strain measurements were were obtained per rosette.
- The strain data were analyzed to determine the state of stress at each station and average stress in the volume of rock encompassing the 14-stations.

Limitations of Surface Relief Methods :

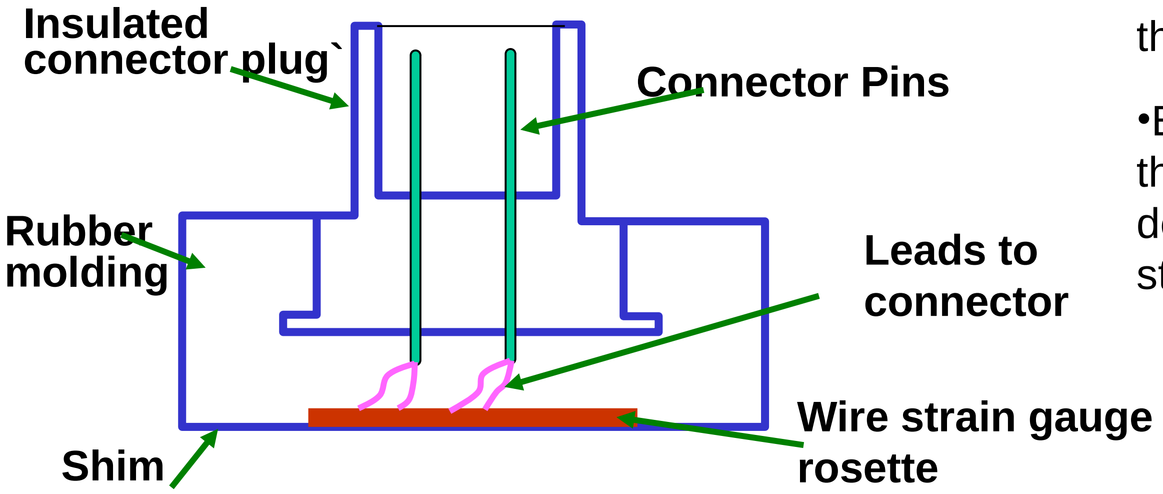
- The gauges and pins are subject to aggressive conditions such as humidity and dust.
- The strains and displacements are measured on a rock that may have been disturbed by the excavation process itself.
- Stress concentration factors have to be assumed to relate the stresses measured in the walls of the excavation to the far field stress components.

Borehole Over Coring Methods :



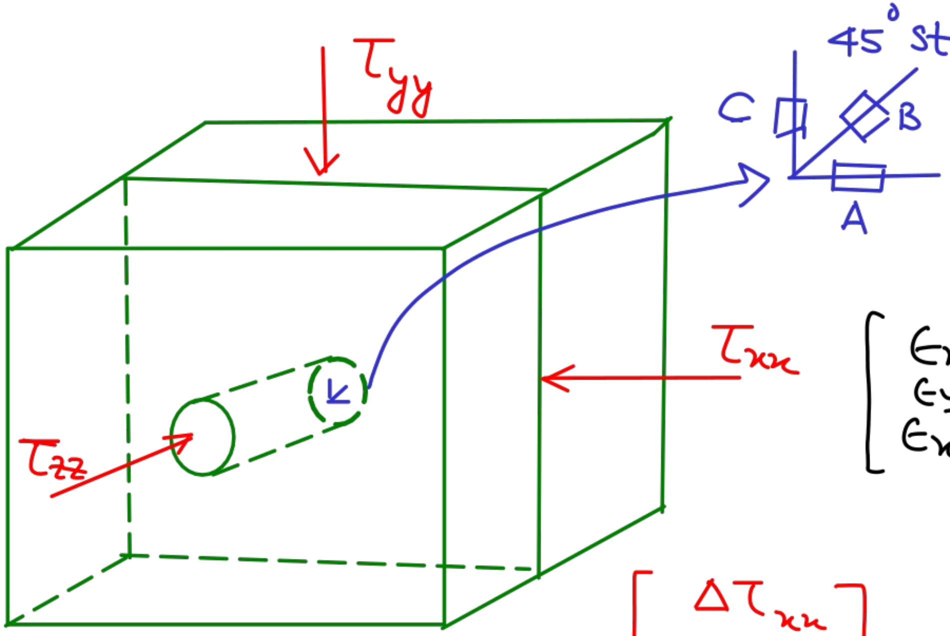
CSIR Doorstopper :-

- The guage is 35mm in dia.
- Suitable for 60 mm BH dia. Or 76mm dia.
- Base consists of a 45^0 strain rosette , cemented on a thin shim.
- The shim is molded in rubber,(protection against dust & water)
- The cell is pushed by compressed air & cemented on a smooth surface at the base of the drill hole.



- Before overcoring the strains are recorded in the strain gages.
- By knowing the elastic properties of the rock, the stresses at the end of the hole can be determined , & are then related to the insitu stress.

Door Stopper Stress Calculations



$$\begin{bmatrix} \epsilon_A \\ \epsilon_B \\ \epsilon_C \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0.5 & 0.5 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \epsilon_{xy} \end{bmatrix}$$

$$\begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \epsilon_{xy} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0.5 & 1 & 0.5 \end{bmatrix} \begin{bmatrix} \epsilon_A \\ \epsilon_B \\ \epsilon_C \end{bmatrix}$$

$$\begin{bmatrix} \Delta\tau_{xx} \\ \Delta\tau_{yy} \\ \Delta\tau_{xy} \end{bmatrix} = \frac{E}{1-\nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & (1-\nu) \end{bmatrix} \begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \epsilon_{xy} \end{bmatrix}$$

Stress components acting at the ends of the borehole.

Laboratory experiments on loaded prisms of Steel/Rock

By Bonnechere (1968) & Van Heerden (1968):–

Work: → effect of Stress Components (τ_{xx}, τ_{yy} & τ_{xy}) in rock upon Stress Components ($\Delta\tau_{xx}$, $\Delta\tau_{yy}$ & $\Delta\tau_{xy}$) acting upon the end of borehole.

$$\begin{bmatrix} \Delta\tau_{xx} \\ \Delta\tau_{yy} \\ \Delta\tau_{xy} \end{bmatrix} = \begin{bmatrix} a & b & c \\ b & a & c \\ 0 & 0 & d \end{bmatrix} \begin{bmatrix} \tau_{xx} \\ \tau_{yy} \\ \tau_{xy} \end{bmatrix}$$

Constants	Bonnechere	Van Heerden (Extensive Study on many rocks)
a	1.25	1.25
b	0	(–)0.064
c	(–)0.75(0.5+ν)	(–)0.75(0.645+ν)
d	1.25	1.25

CSIRO MINDATA Hollow Inclusion Stress Cell

Features:

- Recommended by the ISRM for Stress Determination
- Designed for monitoring during overcoring
- Short or Long term monitoring of Triaxial Stress
- Complete 3D stress tensor from one measurement
- Fully encapsulated electronics
- Range of grouts for specific application temperatures

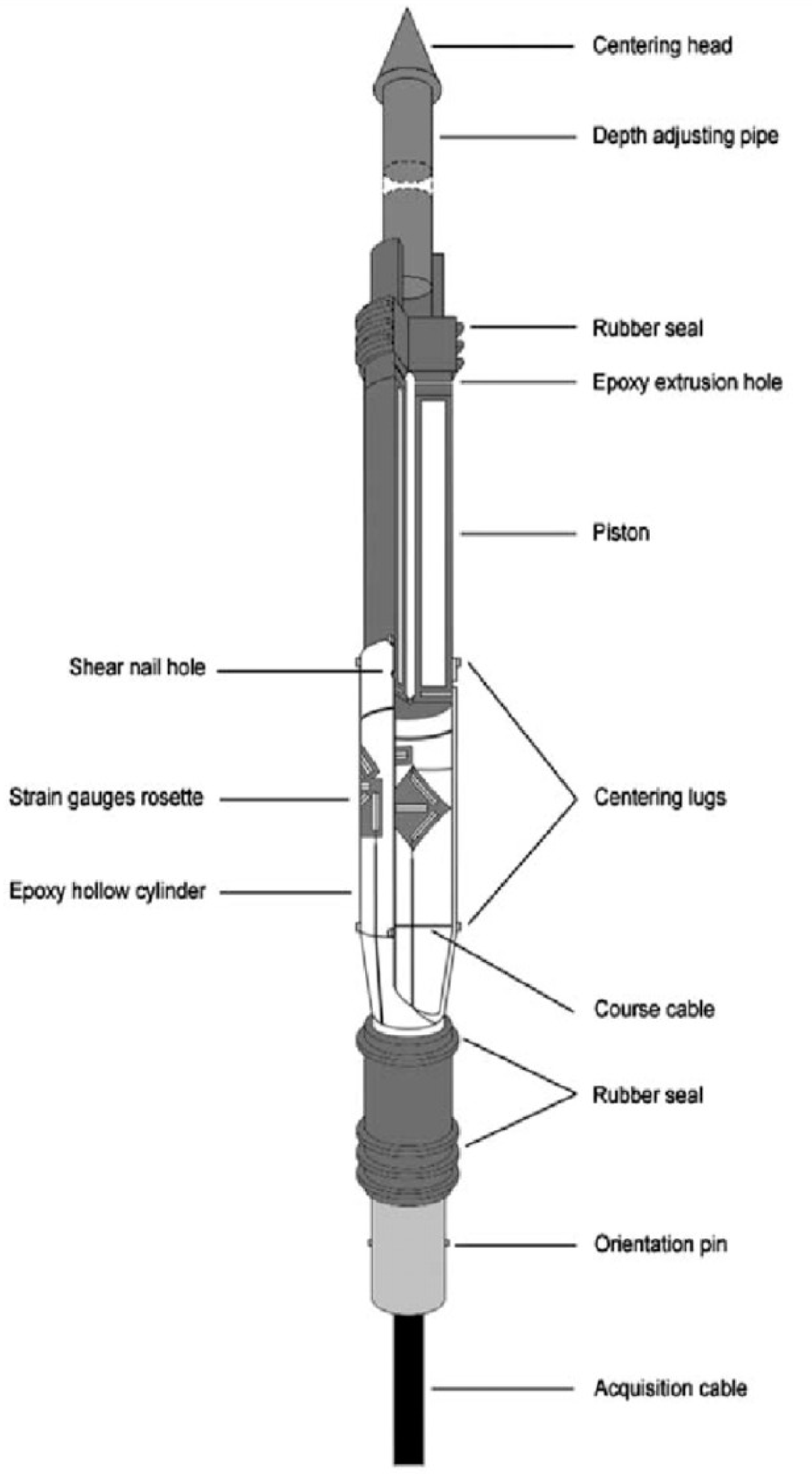
Video Link:

https://youtu.be/AmzEyY_8d0s

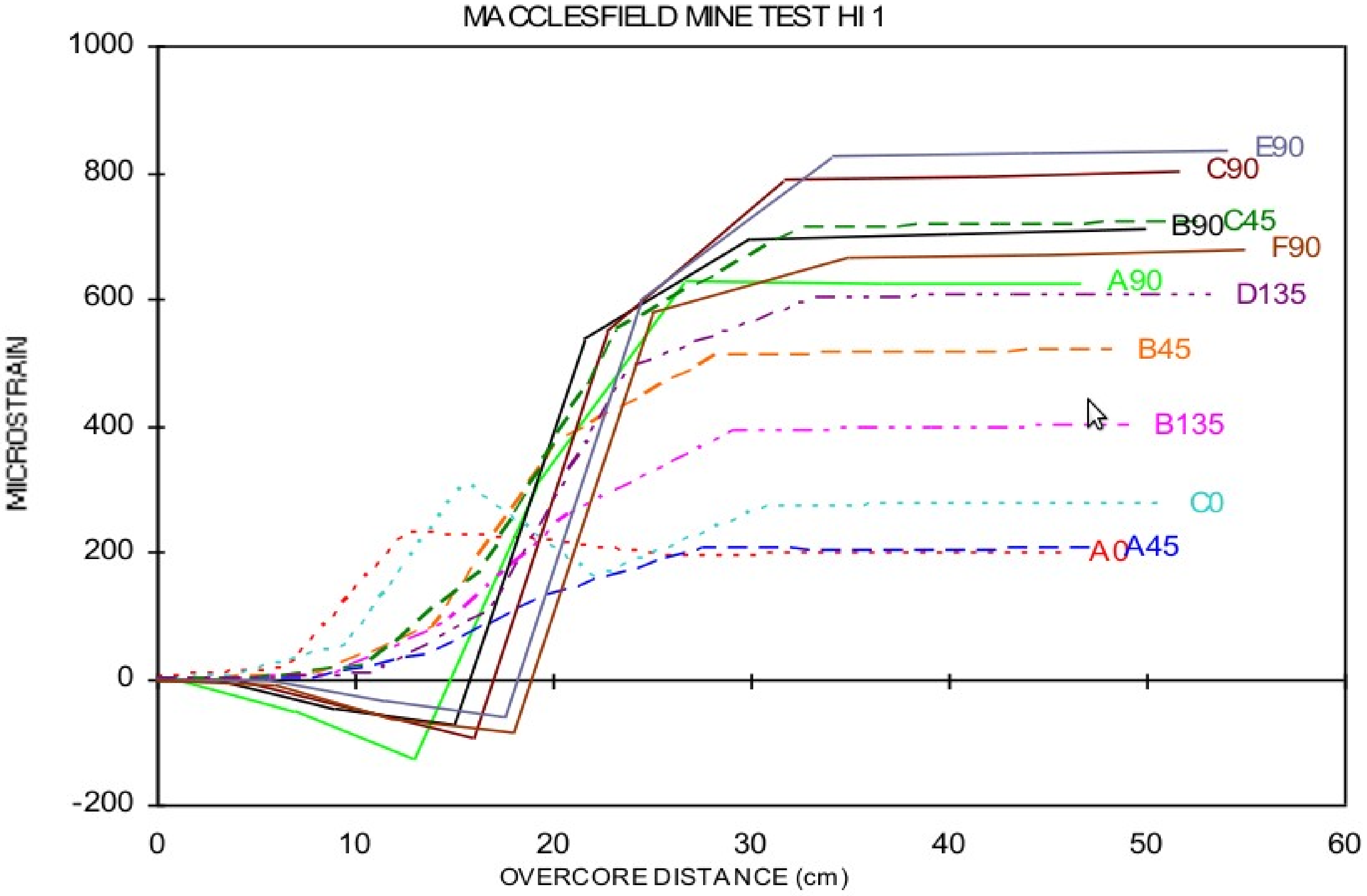
https://youtu.be/jSEAvP5za_8

The HI Cell consists of an array of strain Gauges that are encapsulated in the wall of a hollow pipe with known Elastic Modulus. The cell is epoxy grouted into a bore-hole and monitored for strain response during overcoring or left permanently installed for measuring relative stress over time.

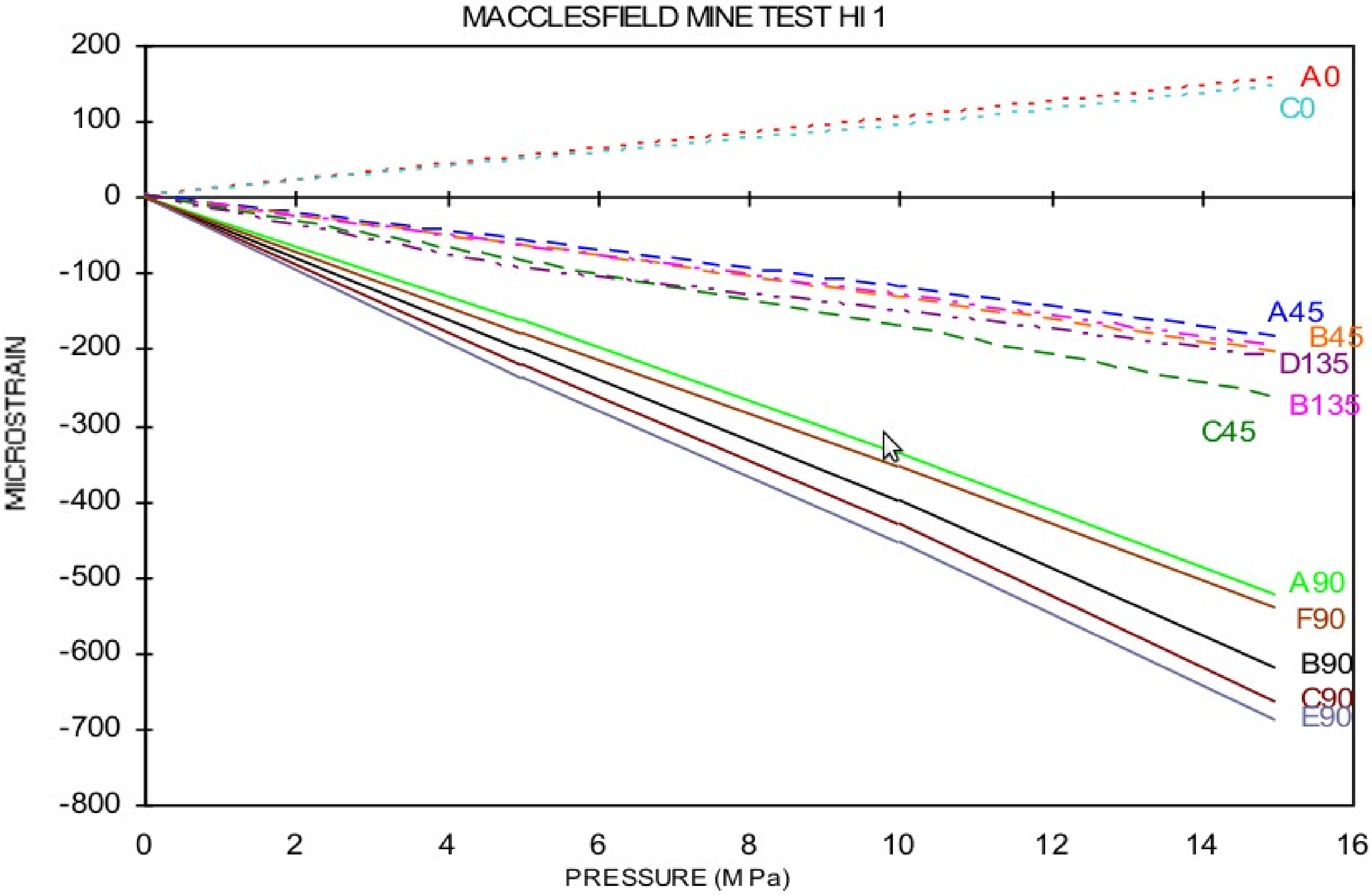
Schematic view of the 12 gauges CSIRO HI Cell



- The HI Cell is installed into a 38mm hole by filling the hollow body of the cell with a pre-formulated epoxy cement.
- Cement is extruded by the piston when the cell reaches target depth.
- Pushing the body of the cell, via installation rods, activates the piston and a trip wire within the cell registers completion of extrusion.
- Multiple rubber seals confine the grout flow to around the cell.
- Prior to installation, the hole is cleaned with compressed air and the walls of the hole prepared for the HI Cell.
- Once the grout has cured, the strain gauges are fully bonded 1.5 to 2.0 mm from the borehole wall, for which allowance is made during data reduction.
- Monitoring of the strain response is possible during overcoring via the data cable, which runs though the drilling rods and a modified water swivel to a strain indicator.
- The ability to record strain gauge information during overcoring provides valuable information, and indicates conditions such as the onset of core cracking, cell de-bonding or inelastic response of the rock.
- **The retrieved overcored section of rock containing the cell may be restressed in a biaxial pressure chamber to derive the actual rock properties of Young's Modulus and Poisson's Ratio.**
- Long term stress change monitoring is possible with the HI Cell and is best suited to monitoring compressive or tensile stress changes in the long term.



MACCLESFIELD MINE TEST HI 1
1000
800
600
400
200
0
-200
MICROSTRAIN
0
10
20
30
40
50
60
OVERCORE DISTANCE (cm)
E90
C90
B90
C45
F90
A90
D135
B45
B135
C0
A0
A45



MACCLESFIELD MINE TEST HI 1
200
100
0
-100
-200
-300
-400
-500
-600
-700
-800
MICROSTRAIN
0
2
4
6
8
10
12
14
16
PRESSURE (M Pa)
A0
C0
A45
B45
D135
B135
C45
A90
F90
B90
C90
E90


GEOKON
Made in the U.S.A.

IS Codes for Stress Measurement

IS 13946 (Part 1) : 1994 :- DETERMINATION OF ROCK STRESS- PART 1 USING THE HYDRAULIC FRACTURING TECHNIQUE

IS 13946 (Part 2) : 1994 :- DETERMINATION OF ROCK STRESS- PART 2 USING A USBM-TYPE GAUGE

IS 13946 (Part 3) : 1994 :- DETERMINATION OF ROCK STRESS- PART 3 USING A CSIR OR CSIRO-TYPE 9 OR 12 STRAIN GAUGES

IS 13946 (Part 4) : 1994 :- DETERMINATION OF ROCK STRESS- CODE OF PRACTICE PART 4 USING FLAT JACK TECHNIQUE